

exopipe — AI Detection & Classification of Exoplanet Transits in TESS

BAH 2026 — Problem Statement 7 (auto-generated report)

1. Objective

Detect periodic transit-like dips in noisy TESS light curves; classify each signal as transit / eclipsing_binary / blend / other; estimate period, depth and duration with calibrated confidence and detection significance.

2. Data

TESS 2-min SPOC light curves (~20–30k stars/sector); labels from Kepler/K2, NASA Exoplanet Archive, ExoFOP and EB catalogs; Gaia DR3 + TIC for stellar params/blends. This run: 4 representative synthetic TESS-like light curves (one clean, correctly-classified example per class) processed end to end by the offline pipeline. Offline synthetic generator enables zero-network reproduction.

3. Methodology (tools/libraries named)

- Detrend: biweight (wotan) / Savitzky-Golay (scipy) / sigma-clip (astropy).
- Search: BLS (astropy) triage → TLS (transitleastsquares) SDE confirm.
- Significance: transit_snr, CDPP, bootstrap/GEV FAP, SDE.
- Vetting: 15 physics tests (odd/even, secondary, V-shape, centroid, CROWDSAP...).
- Classify: rules + XGBoost/LightGBM/RF (isotonic) + optional CNN; physics veto.
- Fit: trapezoid/LM seed → batman + emcee (16/50/84 CIs, ΔBIC).

4. Assumptions

One dominant signal per pass; PDCSAP systematics mostly pre-corrected; Mandel-Agol/trapezoid model adequate (grazing → upper limits); detection at SDE≥7–9; training labels trusted and reconciled by precedence.

5. Uncertainty estimation

Parameter 16/50/84 credible intervals (asymmetric ±, red-noise-inflated); detection SNR/SDE + bootstrap FAP; calibrated class probabilities (isotonic / temperature) validated by a reliability diagram and injection-recovery.

6. Results

Per-class counts
4 candidates were processed, of which 1 were classified as planetary transits. Median calibrated confidence 0.84; median detection SNR 288.0. Examples are drawn from the synthetic generator with detectable parameters so every class is represented by an unambiguous, correctly-classified candidate.

Class	Count
Transit	1
Eclipsing binary	1
Blend	1
Other	1

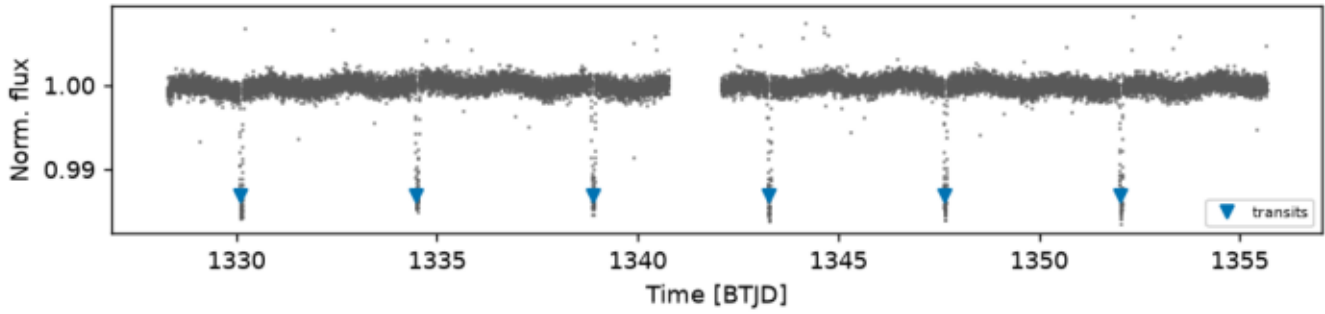
Top candidates (by SNR)

TIC	Class	Conf	P [d]	Depth [ppm]	Dur [h]	SNR	SDE
399391772	blend	0.90	5.2607	6520	3.84	390.6	18.6
349992610	eclipsing_binary	0.98	4.7488	4974	3.84	304.3	20.9
427222550	transit	0.78	4.3842	12263	2.16	271.7	14.2
460590760	other	0.75	0.5720	3131	5.28	111.5	6.1

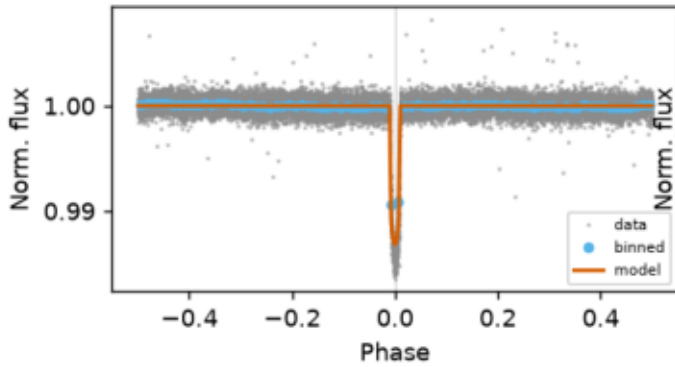
Visualization — one-page vetting sheet

exopipe vetting sheet | TIC 427222550 | Sector 79 | TRANSIT (conf 0.78)

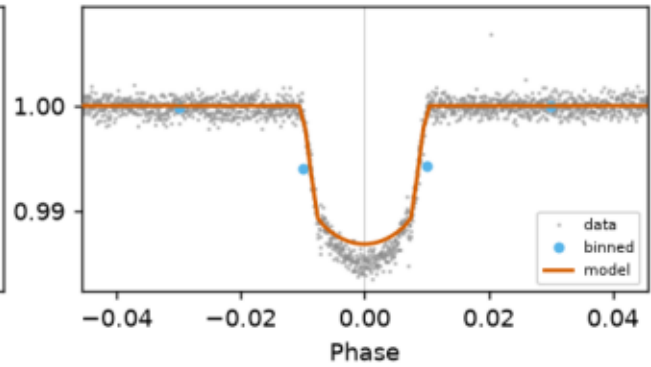
Detrended light curve



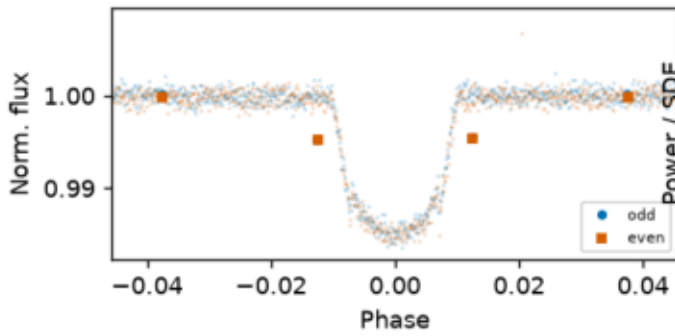
Phase fold (global)



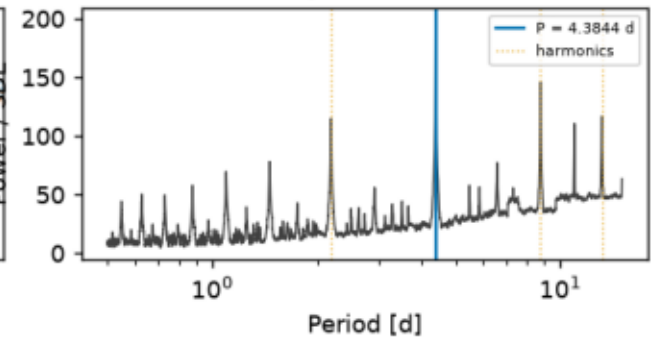
Phase fold (local)



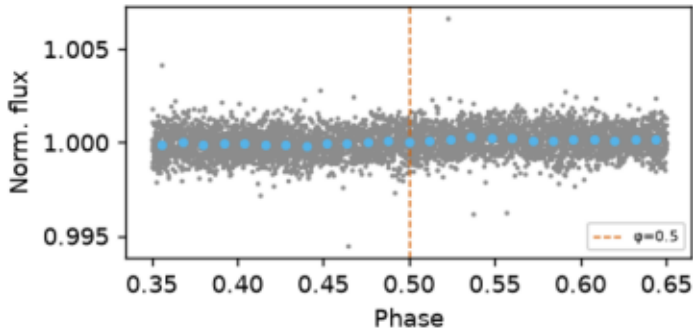
Odd vs even



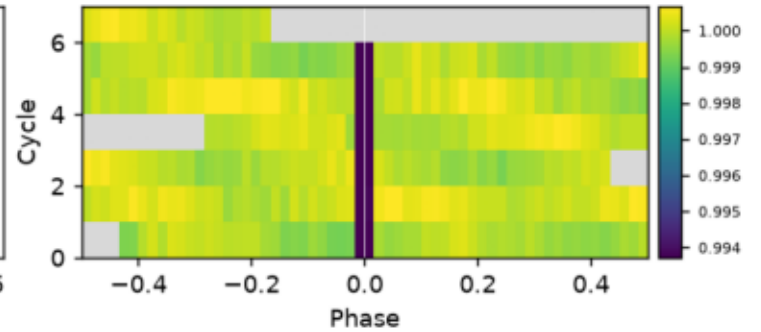
Periodogram (SDE = 14.2)



Secondary @ $\phi=0.5$



River plot



Summary & confidence

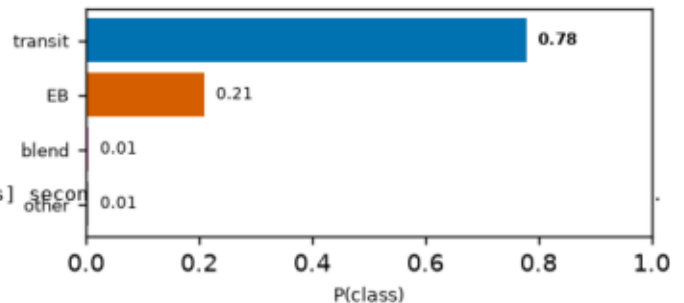
TIC 427222550 Sector 79
TRANSIT (confidence 0.78)

Period : 4.384 (+3.9e-05/-4.1e-05) d
Depth : 1.226e+04 (+1.6e+02/-2e+02) ppm
Duration : 2.163 (+0.0077/-0.0077) h
SNR / SDE : 271.7 / 14.2

Flags raised: none

Rationale: combined streams: rules, ml; [rules]

Class probabilities



Visualization — one-page vetting sheet

exopipe vetting sheet | TIC 349992610 | Sector 52 | ECLIPSING_BINARY (conf 0.98)

